Question: What regular season factors contribute the most to future NBA playoff success, and how can these factors be used to predict a future champion?

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IB SL Analysis and Approaches Internal Assessment

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Introduction:

You may or may not have ever watched an actual basketball game, but have you ever wondered what factors truly contribute to the winning success of a team? What about the factors that contribute to the chance of a team winning the championship? These factors can be determined through the application of statistics, a branch of mathematics that involves the collection, analysis, interpretation, presentation, and organization of data (John, 2019). Because of my fascination and genuine passion for the game of basketball and the subject of math, I decided to investigate the top three factors in determining championship likelihood from the 2017-18 to 2022-23 seasons, and from there, predict the next NBA champion, which will be for the 2023-2024 season as of this exploration.

NBA Background:

The National Basketball Association (NBA) was founded in New York City on June 6, 1946, when it was then called the Basketball Association of America. After merging with the competing National Basketball League (NBL), the Basketball Association of America eventually changed its name to the National Basketball Association on August 3, 1949 (*This Date in the NBA: August*, n.d.).

For years, fans of the NBA have tried to predict which team will make it to the playoffs, which team will make it to the finals, and even which team will take home the gold, winning the NBA championship. There are 30 teams in the NBA and 16 of those teams make it to the playoffs each year: eight from the Eastern conference (teams on the east side of the United States), and eight from the Western conference (teams on the west side of the United States). Two of those 16 teams will go on to compete for the championship. Each team has a "seed", which is their ranking based on their win percentage during the regular season in their respective conference. The top eight seeds from each conference make it to the playoffs, where there are four best-of-seven rounds played between teams, including the finals where teams compete for the championship. The main prediction method involves using the statistics provided by the NBA on each team, such as field goal percentage, three-point percentage, net rating, and so on. Each of these statistics may serve as better predictors than others for predicting the playoff success of a team, making it rational to test them.

Rationale and Relevance:

Through this exploration, I will aim to analyze the validity of using mathematics in predicting championship success while attempting to keep the math as concise as possible. This particular investigation relates back to my specific curriculum in IB Analysis and Approaches because Pearson's coefficient of determination (R^2) will be used to determine the overall strength of each model I will create to find which regular season factors are the best predictors. This is one of the statistical measurements we used when correlating multiple sets of data in various population and sample types.

Hypotheses:

H_0: Of all regular season factors that will be used to predict championship likelihood, wins during the regular season will be the most accurate predictor accounting for at least 60% of the variance in championship likelihood.

H_0: None of the regular season factors will be able to account for more than 50% of the variance in championship likelihood.

Mathematical Background:

Because there are many different factors that can determine the chance of winning an NBA championship, many different models have been created to predict which team will win it. Some models use machine learning, such as the one created by a user named "TheJK" in a blogpost, which correctly predicted the champions of the 2019-20, 2020-21, and 2021-22 seasons (TheJK, 2022). Unlike my investigation, however, this model used many different statistics and past playoff data. My investigation will only use three factors for brevity, each only coming from the regular season so it can be more applicable to each NBA team rather than teams with previous playoff success, that correlate most strongly with playoff success, and attempt to predict the chance of winning an NBA championship. I will also only include the data of teams that made it to the playoffs during the past five seasons, as I believe it is redundant to include teams that have not made it to the playoffs recently in a championship prediction. Furthermore, the only team that has not made the playoffs since the 2017-18 season are the Charlotte Hornets. Their exclusion can be justified by the fact that within the past 21 seasons combined, the team has won a total of just three playoff games (StatMuse, n.d.).

Factors:

I have chosen five factors and will decide by the end of this investigation which three of those five factors are the most important for determining the chance of winning a championship.

Wins During the Regular Reason (WINS RS):

Wins during the regular season (WINS RS) is the total number of wins out of a standard 82 total games that a team plays. I chose this factor because the most important thing a team can do during the regular season to potentially compete for and win the championship is to win as much as possible. Based on prior NBA knowledge, more wins during the regular season generally correlates with playoff success.

Three-Pointers Made During the Regular Season (3PM RS):

Three-pointers made during the regular season (3PM RS) is the total number of three-point shots a team successfully made during the regular season. I chose this because the quantity of three-point shots has dramatically risen over the past decade, with a dramatic increase beginning during the 2013 NBA season.

Assist Percentage During the Regular Season (AST% RS):

Assist percentage during the regular season (AST% RS) measures the percentage of field goals (shots) made off assists out of the total number of field goals a team made during the entire regular season. I chose this statistic because more assists means more passes, and more passes often open up more space on the floor for players to operate and thus score, potentially increasing a team's efficiency.

$$\frac{100 \times AST}{AST + FGA + (0.44 \times FTA) + TO}$$

where
AST = assists
FGA = field goals attempted
FTA = free throws attempted
TO = turnovers

Field Goal Percentage During the Regular Season (FG% RS):

Field goal percentage during the regular season (FG% RS) is the percentage of shots that a team made out of their total shot attempts. I chose this statistic because it is a simple metric used to determine efficiency in basketball, which a team needs if they wish to be successful.

$$\frac{FGM}{FGA}$$

where
FGM = field goals made
FGA = field goals attempted

Net Rating During the Regular Season (NETRTG RS):

Net rating during the regular season (NETRTG RS) is a measure of a team's defensive rating subtracted from their offensive rating per 100 possessions. The number of possessions is the number of times a team secured the ball during the game. Defensive rating is how many points Team A, for example, allows per game, meaning how many points their opponent, Team B, scores per game. Offensive rating is the opposite: how many points Team A scores per game. I chose this statistic because I believe it can be used to determine the strength of a team's wins, meaning how many points a team wins by on average.

$$(100 \times (\frac{PTS}{POSS})) - (100 \times (\frac{OPP\ PTS}{OPP\ POSS}))$$

where
PTS = points scored
POSS = number of possessions
OPP = opponents

Mathematical Concepts:

All of the mathematical concepts that will be used in this investigation involve statistics analysis. The main concepts that I will focus on are Pearson's coefficient of determination (R^2) and multiple linear regression (MLR).

Pearson's Coefficient of Determination (R^2):

Pearson's coefficient of determination, denoted by the notation R^2, can be used to demonstrate exactly how correlated one dependent and one independent variable are (Nevil, 2021). Since the value is squared, the value is always positive, being a decimal between 0 and 1. The closer the R^2 value is to 1, the more variance in the dependent variable can be explained by the independent variable, meaning the more correlated the two variables are. For my investigation, this will allow me to not only examine whether there is a correlation between one statistic and championship success but also how strong that correlation is. However, this concept is limited because it cannot be used for the goodness of fit in nonlinear models (Spiess & Neumeyer, 2010), but for my investigation, this is fine because I will only be using linear models. The formula for R^2 is the following (Tiara, 2016):

$$r^2 = \left(\frac{SS_{xy}}{\sqrt{SS_{x^2} \times SS_{y^2}}}\right)^2$$

where

SS_{xy} = the sum of each x-value minus its average multiplied by the sum of each y-value minus its average

SS_{x^2} = the squared value of the sum of each x-value minus its average

SS_{y^2} = the squared value of the sum of each y-value minus its average

$$SS_{x^2} = \Sigma x^2 - \left(\frac{\Sigma x}{n}\right)^2$$

$$SS_{y^2} = \Sigma y^2 - \left(\frac{\Sigma y}{n}\right)^2$$

$$SS_{xy} = \Sigma xy - \frac{(\Sigma x)(\Sigma y)}{n}$$

Adjusted Coefficient of Determination (Adjusted R^2):

In addition to R^2, there is also an adjusted R^2 measurement which serves the same purpose as R^2, but is a version that accounts for the number of independent variables in the model (Investopedia, 2022). This value is always less than or equal to R^2 (IBM, 2023), but will be the main value focused on throughout this investigation when examining the accuracy of a prediction model specifically with multiple variables, as some of the prediction models I will create will include multiple independent variables. Its formula is the following (Gupta, 2021):

$$Adjusted\ R^2 = 1 - \frac{(1 - R^2)(N - 1)}{N - p - 1}$$

Where

R^2 Sample R-Squared

N Total Sample Size

p Number of independent variable

Multiple Linear Regression (MLR):

Multiple linear regression (MLR) two or more independent variables that attempt to predict the dependent variable rather than just one like single linear regression, for example. This may be more accurate than single linear regression since it is taking more variables into consideration. This is significant for my particular investigation because there are a myriad of factors that contribute to an NBA team's championship likelihood, and thus, including multiple factors into a single equation may increase the accuracy of the prediction. However, multiple linear regression is not good at explaining the relationship between independent and dependent variables if those relationships are not linear, making its use limited. Its formula is the following (Dawson, 2021):

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ \ldots \beta_i X_i + \ \varepsilon$$

where

y = dependent variable

β_i = slope coefficient

X_i = independent variable

β_0 = y-intercept

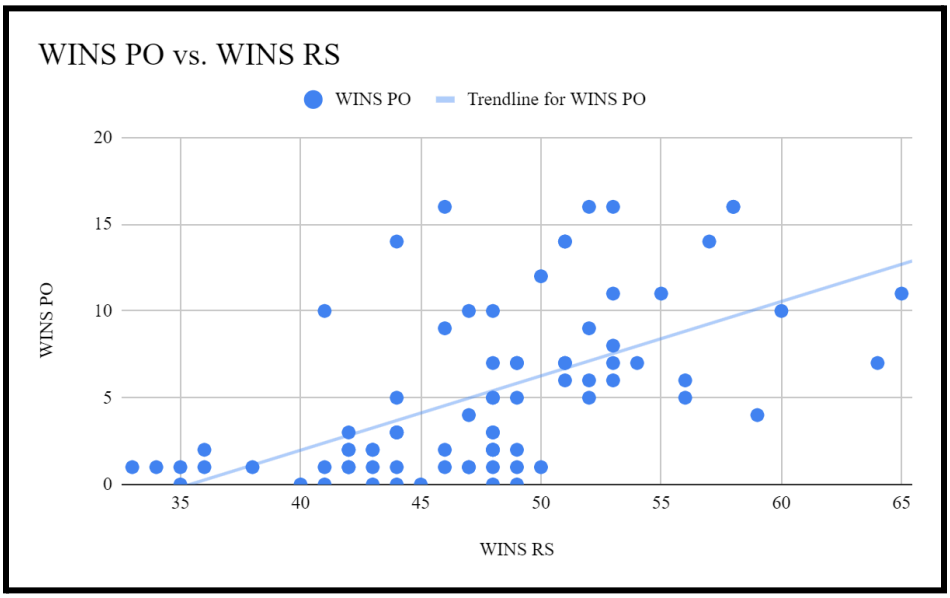
ε = error term

Exploration:

To begin, I will find the values for each of the five regular season statistics I chose for each playoff team from the 2017-18 season to the 2021-22 season. Championship success will be the dependent variable located on the Y-axis and will be measured as total playoff wins, which a team needs 16 during a playoff run of to win the championship. Each of the regular season statistics I chose will be the independent variables located on the X-axis.

Calculations and Graphs:

WINS RS:



Graph created by Google Sheets containing all WINS RS values against playoff wins for all playoff teams from the 2017-18 to 2021-22 NBA seasons.

All x-values (WINS RS):
[49, 44, 47, 47, 43, 48, 44, 48, 59, 48, 48, 52, 55, 65, 50, 58, 41, 48, 42, 42, 49, 50, 48, 48, 49, 53, 51, 54, 53, 60, 57, 58, 35, 43, 45, 33, 35, 43, 44, 44, 44, 56, 49, 53, 46, 48, 44, 52, 40, 34, 36, 38, 41, 42, 42, 42, 47, 52, 48, 49, 41, 47, 51, 46, 44, 43, 46, 48, 36, 46, 48, 49, 51, 56, 51, 64, 52, 53, 51, 53]

All y-values (WINS PO):
[0, 1, 1, 1, 2, 2, 3, 3, 4, 5, 5, 5, 11, 11, 12, 16, 0, 0, 1, 1, 1, 1, 2, 3, 5, 6, 7, 7, 8, 10, 14, 16, 0, 0, 0, 1, 1, 2, 3, 3, 5, 5, 7, 7, 9, 10, 14, 16, 0, 1, 1, 1, 1, 2, 2, 3, 4, 6, 7, 7, 10, 10, 14, 16, 0, 1, 1, 1, 2, 2, 2, 2, 6, 6, 7, 7, 9, 11, 14, 16]

First, I will calculate SS_{x^2}, which I will begin by calculating the value of Σx^2.

$$\Sigma x^2 = (49^2) + (44^2) + (47^2) + \ldots = 184675$$

Then $(\Sigma x)^2$.

$$(\Sigma x)^2 = ((49) + (44) + (47) + \ldots)^2 = 3809^2 = 14508481$$

Then, I will multiply this value by $\frac{1}{80}$ since my sample count is 80 (16 teams over 5 seasons).

$$14508481 \times \frac{1}{80} = 181356.0125$$

Finally, I will subtract this value from Σx^2.

$$184675 - 181356.0125 = SS_{x^2} = 3318.9875$$

Next, I will calculate SS_{y^2}, which is the same process except I will be using the y-values.

$$SS_{y^2} = 1830.4875$$

Next, I will calculate SS_{xy}, which I will begin by finding Σxy.

$$\Sigma xy = (49 \times 0) + (44 \times 1) + (47 \times 1) + \ldots = 21372$$

Then, $(\Sigma x)(\Sigma y)$.

$$(\Sigma x)(\Sigma y) = (49 + 44 + 47 + \ldots) \times (0 + 1 + 1 + \ldots) = 3809 \times 419 = 1595971$$

Then, I will multiply this value by $\frac{1}{80}$ since my sample count is 80.

$$1595971 \times \frac{1}{80} = 19949.6375$$

Finally, I will subtract this value from Σxy.

$$21372 - 19949.6375 = SS_{xy} = 1422.3625$$

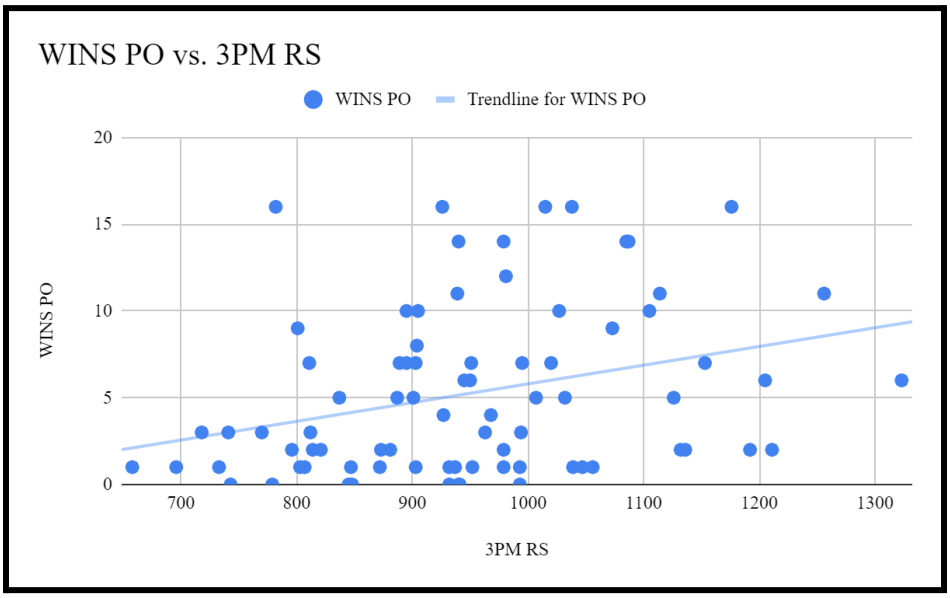
To calculate the final value for R^2, I will now plug each of these numbers into the equation $r^2 = \left(\frac{SS_{xy}}{\sqrt{SS_{x^2} \times SS_{y^2}}}\right)^2$.

$$r^2 = \left(\frac{1422.3625}{\sqrt{3318.9875 \times 1830.4875}}\right)^2 = (0.57706415923)^2 \approx 0.333$$

rounded to 3.s.f

An average R^2 value of 0.333 for the number of playoff wins versus the number of regular season wins indicates that about 33.3% of the variance is able to be explained by the number of regular season wins a team has.

3PM RS:



Graph created by Google Sheets containing all 3PM RS values against playoff wins for all playoff teams from the 2017-18 to 2021-22 NBA seasons.

All x-values (3PM RS):
[845, 903, 658, 696, 814, 881, 718, 741, 968, 837, 887, 901, 939, 1256, 981, 926, 993, 779, 1047, 937, 932, 993, 821, 812, 1032, 1323, 889, 903, 904, 1105, 1087, 1015, 941, 848, 743, 807, 952, 1136, 770, 963, 1126, 1007, 895, 995, 801, 905, 979, 782, 932, 733, 979, 803, 847, 796, 1132, 994, 927, 1205, 1020, 811, 895, 1027, 940, 1038, 940, 1056, 872, 1039, 873, 1211, 979, 1192, 950, 945, 1153, 951, 1073, 1114, 1085, 1176]

All y-values (WINS PO):
[0, 1, 1, 1, 2, 2, 3, 3, 4, 5, 5, 5, 11, 11, 12, 16, 0, 0, 1, 1, 1, 1, 2, 3, 5, 6, 7, 7, 8, 10, 14, 16, 0, 0, 0, 1, 1, 2, 3, 3, 5, 5, 7, 7, 9, 10, 14, 16, 0, 1, 1, 1, 1, 2, 2, 3, 4, 6, 7, 7, 10, 10, 14, 16, 0, 1, 1, 1, 2, 2, 2, 2, 6, 6, 7, 7, 9, 11, 14, 16]

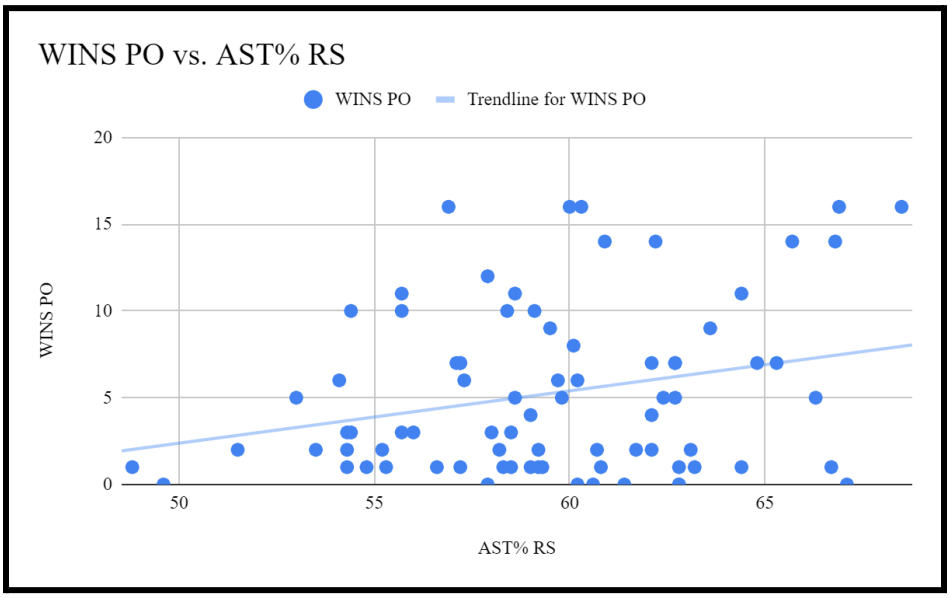
The same math shown in the 'WINS RS' section will be used to calculate the R^2 value for playoff wins versus three-pointers made during the regular season.

$$r^2 = 0.0933$$

rounded to 3.s.f

An average R^2 value of 0.0933 for the number of playoff wins versus the number of three-pointers made during the regular season means that three-pointers were able to explain less than 10% of the variance in playoff wins.

AST% RS:



Graph created by Google Sheets containing all AST% RS values against playoff wins for all playoff teams from the 2017-18 to 2021-22 NBA seasons.

All x-values (AST% RS):

[49.6, 58.5, 55.3, 58.3, 63.1, 53.5, 58.5, 54.4, 59, 62.7, 58.6, 66.3, 58.6, 55.7, 57.9, 68.5, 57.9, 62.8, 59.2, 63.2, 54.8, 64.4, 58.2, 58, 62.4, 54.1, 64.8, 65.3, 60.1, 54.4, 66.8, 60.3, 60.6, 62.8, 61.4, 60.8, 48.8, 59.2, 54.3, 56, 53, 59.8, 57.1, 62.7, 63.6, 55.7, 65.7, 60, 67.1, 59, 56.6, 62.8, 54.3, 60.7, 51.5, 55.7, 62.1, 57.3, 62.1, 57.2, 59.1, 58.4, 62.2, 56.9, 60.2, 59.3, 57.2, 66.7, 62.1, 61.7, 54.3, 55.2, 60.2, 59.7, 57.2, 62.7, 59.5, 64.4, 60.9, 66.9]

All y-values (WINS PO):

[0, 1, 1, 1, 2, 2, 3, 3, 4, 5, 5, 5, 11, 11, 12, 16, 0, 0, 1, 1, 1, 1, 2, 3, 5, 6, 7, 7, 8, 10, 14, 16, 0, 0, 0, 1, 1, 2, 3, 3, 5, 5, 7, 7, 9, 10, 14, 16, 0, 1, 1, 1, 1, 2, 2, 3, 4, 6, 7, 7, 10, 10, 14, 16, 0, 1, 1, 1, 2, 2, 2, 2, 6, 6, 7, 7, 9, 11, 14, 16]

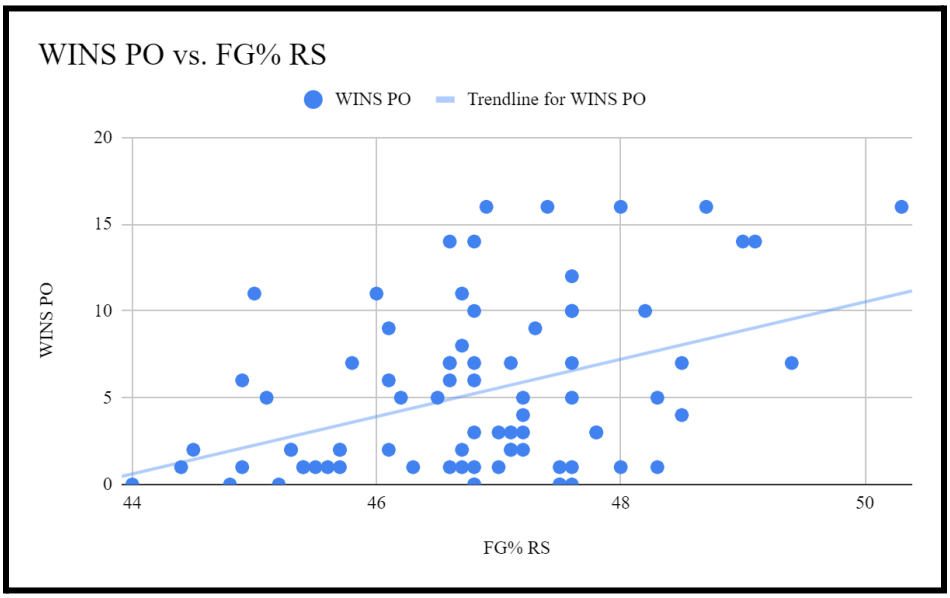
The same math shown in the 'WINS RS' section will be used to calculate the R^2 value for playoff wins versus assist percentage during the regular season.

$$r^2 = 0.0666$$

rounded to 3.s.f

An average R^2 value of 0.0666 for the number of playoff wins versus the assist percentage a team had during the regular season means that less than 7% of the variance in the number of playoff wins a team had could be explained by their assist percentage during the regular season.

FG% RS:



Graph created by Google Sheets containing all FG% RS values against playoff wins for all playoff teams from the 2017-18 to 2021-22 NBA seasons.

All x-values (FG% RS):

[45.2, 45.5, 47.6, 45.7, 46.7, 45.3, 47.8, 47.2, 47.2, 48.3, 46.2, 47.2, 45, 46, 47.6, 50.3, 44, 47.5, 44.9, 45.4, 45.4, 46.8, 47.1, 47.8, 46.5, 44.9, 47.1, 46.6, 46.7, 47.6, 49.1, 47.4, 44.8, 46.8, 47.6, 44.4, 46.3, 46.1, 46.8, 47.1, 45.1, 47.6, 46.6, 45.8, 47.3, 47.6, 46.8, 48, 46.8, 47.5, 46.6, 46.7, 45.6, 47.2, 45.3, 47, 48.5, 46.8, 49.4, 47.6, 46.8, 48.2, 49, 48.7, 47.5, 47, 48, 48.3, 45.7, 45.7, 44.5, 47.1, 46.6, 46.1, 46.8, 48.5, 46.1, 46.7, 46.6, 46.9]

All y-values (WINS PO):

[0, 1, 1, 1, 2, 2, 3, 3, 4, 5, 5, 5, 11, 11, 12, 16, 0, 0, 1, 1, 1, 1, 2, 3, 5, 6, 7, 7, 8, 10, 14, 16, 0, 0, 0, 1, 1, 2, 3, 3, 5, 5, 7, 7, 9, 10, 14, 16, 0, 1, 1, 1, 1, 2, 2, 3, 4, 6, 7, 7, 10, 10, 14, 16, 0, 1, 1, 1, 2, 2, 2, 2, 6, 6, 7, 7, 9, 11, 14, 16]

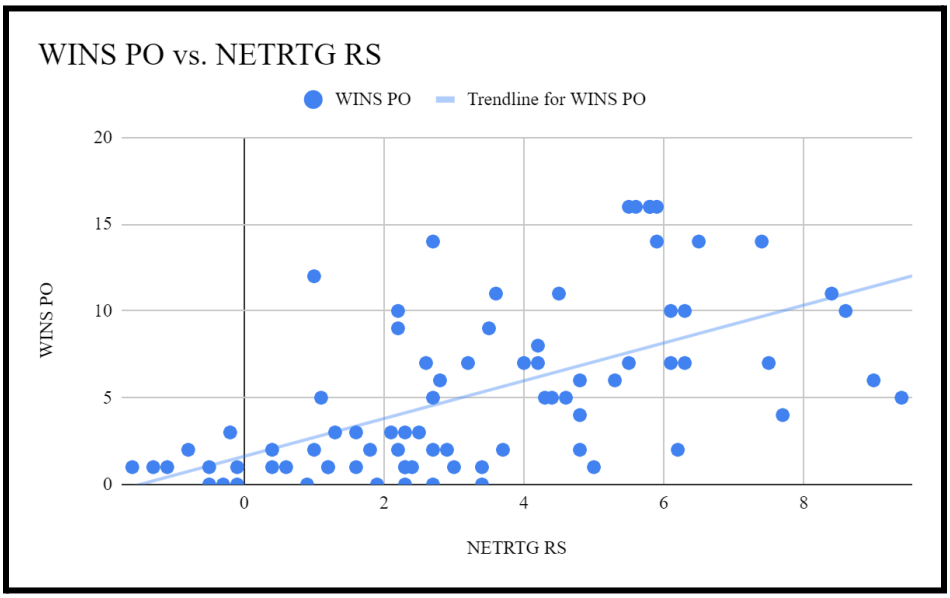
The same math shown in the 'WINS RS' section will be used to calculate the R^2 value for playoff wins versus field goal percentage during the regular season.

$$r^2 = 0.172$$

rounded to 3.s.f

An average R^2 value of 0.172 for the number of playoff wins a team had versus their field goal percentage during the regular season, means that 17% of the variance in playoff wins could be explained by a team's field goal percentage during the regular season.

NETRTG RS:



Graph created by Google Sheets containing all NETRTG RS values against playoff wins for all playoff teams from the 2017-18 to 2021-22 NBA seasons.

All x-values (NETRTG RS):
[2.7, 0.4, 2.3, 3, 0.4, 3.7, -0.2, 1.3, 7.7, 1.1, 4.4, 4.6, 3.6, 8.4, 1, 5.9, -0.3, 3.4, -0.1, 0.6, 3.4, 5, 1, 1.6, 4.3, 4.8, 2.6, 4, 4.2, 8.6, 6.5, 5.8, -0.5, 2.3, 1.9, -1.3, -1.1, 4.8, 2.1, 2.5, 2.7, 9.4, 6.3, 6.1, 2.2, 6.3, 2.7, 5.6, -0.1, -1.6, 1.2, 1.2, 2.4, 2.9, 1.8, 2.3, 4.8, 9, 4.2, 5.5, 2.2, 6.1, 5.9, 5.8, 0.9, 1.6, -0.5, 2.3, -0.8, 2.7, 2.2, 6.2, 2.8, 5.3, 3.2, 7.5, 3.5, 4.5, 7.4, 5.5]

All y-values (WINS PO):
[0, 1, 1, 1, 2, 2, 3, 3, 4, 5, 5, 5, 11, 11, 12, 16, 0, 0, 1, 1, 1, 1, 2, 3, 5, 6, 7, 7, 8, 10, 14, 16, 0, 0, 0, 1, 1, 2, 3, 3, 5, 5, 7, 7, 9, 10, 14, 16, 0, 1, 1, 1, 1, 2, 2, 3, 4, 6, 7, 7, 10, 10, 14, 16, 0, 1, 1, 1, 2, 2, 2, 2, 6, 6, 7, 7, 9, 11, 14, 16]

The same math shown in the 'WINS RS' section will be used to calculate the R^2 value for playoff wins versus net rating during the regular season.

$$r^2 = 0.344$$

rounded to 3.s.f

An average R^2 value of 0.344 for the number of playoff wins a team had versus their net rating during the regular season indicates that a team's net rating during the regular season was able to explain about 36% of the variance in playoff wins.

Individual Seasons:

All R^2 values for each individual season, from 2017-18 to 2021-22, for each statistic were calculated in the same way demonstrated in the 'WINS RS' section above. All information was originally organized using Google Sheets.

	WINS RS	3PM RS	AST% RS	FG% RS	NETRTG RS
R^2 Values 2017-18	0.458	0.380	0.211	0.198	0.184
R^2 Values 2018-19	0.723	0.126	0.0270	0.326	0.506
R^2 Values 2019-20	0.318	0.00200	0.0690	0.210	0.238
R^2 Values 2020-21	0.417	0.0880	0.00	0.420	0.437
R^2 Values 2021-22	0.304	0.146	0.170	0.00200	0.440
Averages (R^2)	0.444	0.148	0.0954	0.231	0.361

Table containing all R^2 values from each individual season to find the averages from the 2017-18 to 2021-22 season.

Factors That Will Be Used in Prediction:

After looking over all of the data, it is clear that net rating, wins, and field goal percentage during the regular season, in that order, are the three most important factors in determining future playoff success for NBA teams out of the original five factors that I selected.

Prediction:

In order to make a prediction, I will make two different models: multiple linear regression using statistics from just one season and multiple linear regression using data from all seasons. Since I plan to use one model for the final prediction, I will test for the accuracy of each equation by finding which equation has the highest coefficient of determination, and use that one to "predict" the 2022-23 NBA champion, which were the Denver Nuggets, based on the three most important factors for playoff success that I calculated. The Sacramento Kings will be excluded from this preliminary prediction as they were a poor NBA playoff team similar to the Charlotte Hornets prior to the 2022-23 NBA season; the Hornets will be excluded in both. Regardless of whether or not a team made it to the playoffs within the past five seasons, I will include the total number of each statistic for each team in an attempt to avoid a methodological bias. Using their average net rating value, number of wins, and field goal percentage regardless of whether or not a team made it to the playoffs within this five-year period will reduce the chance that they will be selected solely based off of their previous playoff success. For example, the Warriors went from winning 57 games in the 2018-19 season and competing in the NBA finals to just 15 during the following season due to unforeseen circumstances such as injuries, but since they made it to the playoffs at least one time during this time span, I will still include a total amount of their statistics for each season.

MLR Equations:

Accuracy for both multiple linear regression equations was calculated based on their respective adjusted coefficient of determination values. Each equation and adjusted R^2 value was derived from the 'XLMiner Analysis ToolPak' extension on Google Sheets.

Individual season data: in 2017-18, the adjusted R^2 value was about 0.614; in 2018-19, it was about 0.699; in 2019-20, it was about 0.171; in 2020-21, it was about 0.499; and in 2021-22, it was about 0.334. In the context of my investigation, this makes the 2018-19 seasons' data the most accurate for explaining variance in the number of playoff wins a team had.

R Square	0.7593631094
Adjusted R Square	0.6992038868

MLR Coefficients of Determination using the 2018-19 season data only (screenshot from Google Sheets).

	Coefficients
Intercept	-49.42049883
X1 - WINS RS	1.163220488
X2 - FG% RS	-0.008407941521
X3 - NETRTG RS	-0.9941486517

MLR Equation Coefficients using the 2018-19 season's data only (screenshot from Google Sheets).

Multiple season data: when using all data from 2017-18 to 2021-22 in the same multiple linear regression equation, the adjusted R^2 value decreases to 0.391, making it relatively inaccurate for explaining the variance in playoff wins.

R Square	0.4259903028
Adjusted R Square	0.4033320253

MLR Correlation Coefficients using all data from the 2017-18 to 2021-22 seasons (screenshot from Google Sheets).

	Coefficients
Intercept	-49.56370291
X1 - WINS RS	0.1950651809
X2 - FG% RS	0.9327598412
X3 - NETRTG RS	0.5600408024

MLR Equation Coefficients using all data from the 2017-18 to 2021-22 seasons (screenshot from Google Sheets).

Based on these accuracy values, I will choose the multiple linear regression equation using the 2018-19 season's data to predict the 2022-23 and 2023-24 NBA champions since it can explain just about 70% of the variance. The exact equation I will use, which was created from Google sheets, rounded to three significant figures is $1.16(X_1) - 0.00841(X_2) - 0.994(X_3) - 49.4$, where X_1 is wins, X_2 is field goal percentage, and X_3 is net rating, all during the regular season.

Preliminary Prediction (2022-23 NBA Champion: The Denver Nuggets):

After substituting every single team's average value for each statistic into this equation, the top three teams in terms of the higher number of predicted playoff wins were the Milwaukee Bucks, the Toronto Raptors, and the Philadelphia 76ers.

$$\text{Milwaukee Bucks: } 1.16(51.4) - 0.00841(5.4) - 0.994(47.7) - 49.4 \approx -37.2$$

$$\text{Toronto Raptors: } 1.16(49) - 0.00841(4.5) - 0.994(45.9) - 49.4 \approx -38.2$$

$$\text{Philadelphia 76ers: } 1.16(49.2) - 0.00841(3.5) - 0.994(47) - 49.4 \approx -39.1$$

each rounded to 3.s.f

The real 2022-23 champions, the Denver Nuggets, were ranked sixth out of 28 total teams, making it around 81% accurate – my calculation for that being the total number of teams below the Nuggets divided by one less than the total number of teams (27). I chose to calculate this accuracy in this way because if the Nuggets were predicted to win the championship for this season, the accuracy would come out to be 100%, which would be correct.

$$Accuracy = \frac{28-6}{28-1} = \frac{22}{27} = 0.815$$

rounded to 3.s.f

	AVG WINS RS	AVG NETRTG RS	AVG FG% RS	# OF WINS PO	
ALL TEAMS					
Timberwolves	34.2	-1	45.6	-55.04599	
Spurs	38.8	0.5	46.7	-50.816005	
Wizards	33.8	-2.4	46.8	-56.691016	
Thunder	37.4	-1.9	44.9	-50.630621	
Bucks	51.4	5.4	47.7	-37.235214	RANK 1
Pacers	40	0.6	47.2	-49.921846	
Raptors	49	4.5	45.9	-38.222445	RANK 2
Pelicans	35.6	-0.5	47.1	-54.917195	
Jazz	48.6	5.5	46.8	-39.589455	
76ers	49.2	3.5	47	-39.075435	RANK 3
Celtics	47.8	4.7	46.2	-39.914327	
Rockets	39.8	0.1	45.2	-48.161641	
Cavaliers	30.8	-4.4	46	-59.358996	
Warriors	44.4	2.6	47.5	-45.132866	
Blazers	41.2	-0.3	45.6	-46.931877	
Heat	44	1.6	46.1	-44.196856	
Pistons	28.6	-3.2	44.5	-60.430088	
Nets	39.4	0.1	46.1	-49.520241	
Magic	28.6	-4.5	44.3	-60.220355	
Clippers	45.6	2.5	47	-43.243025	
Nuggets*	**48.2**	**3**	**47.5**	**-40.72823**	**RANK 6**
Mavericks	38.8	1.2	45.6	-49.728492	
Lakers	39.8	0.2	47	-49.951682	
Grizzlies	36.6	-0.7	45.8	-52.463313	
Knicks	29	-3.5	44.7	-60.162365	
Hawks	31.4	-3	45.7	-58.37657	
Suns	37.8	-1.2	46.8	-52.061108	
Bulls	29.6	-4.1	45.8	-60.554719	
NOT INVOLVED					
Hornets					
Kings					

Table containing all teams that were included in the preliminary prediction (all playoff teams from the 2017-18 to the 2021-22 seasons), as well as all of their respective average number of wins, net rating, and field goal percentage during the regular season, and their subsequent number of wins. Ranking for the top three teams, as well as the champion, was included.

Assessment:

I believe that, considering I gave myself just three factors to create a prediction model with, this is a decent prediction. The Bucks and 76ers are current playoff teams that have been competing for the championship for many consecutive years, and the Raptors had won the championship during the 2018-19 season. However, this could be a methodological bias since I used the multiple linear regression model I created from the 2018-19 data set. The Bucks being predicted to win the NBA championship during the 2022-23 season is rational considering they have been a playoff contender since the 2017-18 season, and had the best record in the league during the 2022-23 season. However, they unexpectedly fell apart during the first round of the playoffs, losing 4 - 1 to the 8th-seeded Miami Heat. According to this model, however, while the order of the teams was predicted with decent accuracy, each of these teams was predicted to have a negative number of playoff wins, but I will still utilize the order of prediction from largest to smallest value.

Primary Prediction (2023-24 NBA Champion: The Milwaukee Bucks):

After substituting every single team's average value for each statistic from the 2018-19 to the 2022-23 seasons into this equation, the top three teams in terms of playoff wins predicted were the Milwaukee Bucks, Philadelphia 76ers, and Denver Nuggets.

$$\text{Milwaukee Bucks: } 1.16(54.2) - 0.00841(6.1) - 0.994(47.6) - 49.4 \approx -33.9$$

$$\text{Philadelphia 76ers: } 1.16(49.6) - 0.00841(3.5) - 0.994(47.4) - 49.4 \approx -39.0$$

$$\text{Denver Nuggets: } 1.16(49.6) - 0.00841(3.4) - 0.994(47) - 48.2 \approx -39.8$$

each rounded to 3.s.f

The Bucks and 76ers were once again both in the top three, which is reasonable since they have both been either close to winning or won a championship numerous times within the past five to six years. The Nuggets moved up to the top three as well, which may be due to their recent championship run. Considering the last prediction was about 81% accurate, this prediction can be seen as relatively strong.

	AVG WINS RS	AVG NETRTG RS	AVG FG% RS	# OF WINS PO	
ALL TEAMS					
Timberwolves	33.2	-1.5	45.9	-56.499985	
Spurs	33.8	-2.2	46.9	-56.792098	
Wizards	32.2	-2.7	47.1	-58.842693	
Thunder	35.8	-2.3	45.1	-52.682057	
Bucks	**54.2**	**6.1**	**47.6**	**-33.893701**	**RANK 1**
Pacers	37.4	-0.3	47.1	-52.830877	
Raptors	45.4	3.2	45.7	-42.188712	
Pelicans	34.4	-0.4	47	-56.210636	
Jazz	46.4	4.4	47	-42.331004	
76ers	49.6	3.5	47.4	-39.009035	RANK 2
Celtics	48.2	5.3	46.7	-39.952373	
Rockets	31.2	-3.3	45.1	-58.009647	
Cavaliers	31	-3.5	46.2	-59.333365	
Warriors	41.6	1.7	47	-47.876297	
Blazers	38	-1.7	46	-51.029703	
Heat	44	1.4	46.2	-44.294574	
Pistons	24.2	-5	44.6	-65.61835	
Nets	42.8	1	47	-46.47841	
Magic	30.4	-4	44.6	-58.43476	
Clippers	46	2.6	47.1	-42.879266	
Nuggets	49.6	3.4	48.2	-39.803394	RANK 3
Mavericks	41.6	1.8	46.3	-47.181338	
Lakers	41.4	0.7	47.4	-48.497487	
Grizzlies	42.4	1.5	46.4	-46.350215	
Knicks	32.6	-2.1	44.9	-56.196939	
Hawks	34.8	-1.8	46.4	-55.138462	
Suns	42.6	1.2	47.4	-47.109692	
Bulls	32.2	-2.3	47	-58.746657	
Kings	35.8	-1.9	47.2	-54.772821	
NOT INVOLVED					
Hornets					

Table containing all teams that were included in the primary prediction (all playoff teams from the 2018-19 to the 2022-23 seasons), as well as all of their respective average number of wins, net rating, and field goal percentage during the regular season, and their subsequent number of wins. Ranking for the top three teams, as well as the champion, was included.

Analysis:

Addressing My Hypothesis:

In my hypothesis, I predicted that wins during the regular season would be the factor that correlates most with playoff wins out of the variables I chose, being able to explain at least 60% of the variance in playoff wins. According to my data, however, my hypothesis must be rejected and the null hypothesis retained. There were no variables that were able to explain more than 50% of the variance in playoff wins. Rather than explaining around 60% of the variance, wins during the regular season was only able to explain around 33.3% of the variance. Instead, net rating during the regular season was the most influential factor out of the ones I chose, explaining around 34.4% of the variance in playoff wins.

Net rating being the most influential factor out of the original ones I chose is reasonable because net rating essentially evaluates the strength of a team's win, not just the fact that a team won. Since having more wins does not always mean that a team will have more success in the playoffs, it makes sense for net rating to be used in addition to the number of wins to determine playoff success. For example, Team A wins by one point on average, whereas Team B wins by five points on average. Team B may be better than Team A because it wins by more points on average, however, this is not taken into account in the wins statistic, making net rating a valuable metric for success.

Other Results:

Wins and Field Goal Percentage:

Wins and field goal percentage during the regular season being the second and third most influential variable out of the original factors is reasonable because the more a team wins during the regular season, the more likely they will be able to win during the playoffs against the same teams since they have more experience being successful.

Field goal percentage being the third strongest variable is also reasonable since it determines the efficiency of a team overall. More efficient teams have higher potential to score within a game, increasing the likelihood of winning.

However, it is important to note that, while in the context of the original factors I chose for predicting championship likelihood these factors were the second and third most influential, none of these variables were are strong enough to be considered significantly impactful in explaining the variance in the number of playoff wins, as none of the R^2 values every exceeded 0.5.

Three-Pointers Made and Average Assist Percentage:

There was overall a downward trend in the three-pointer's significance from 2017-18 to 2021-22, with regular R^2 values dropping from 0.380 to 0.146, with the lowest value coming from the 2019-20 season of just 0.00200, explaining nearly 0% of the variance. Though three-pointers have shown a surge in attempts over the past decade, this decline is reasonable because an increase in quantity correlates with a decrease in value. For

example, during the 2017-18 season, there was only one team that had more than 1,000 three-pointers, whereas this number rose to nine total teams during the 2021-22 season.

Assist percentage during the regular season appeared to be the least significant in determining playoff winning success, with an R^2 value of 0.0666. Because each team uses a different offensive scheme which may involve varying amounts of passing, it is reasonable that other factors may play a larger role in determining playoff success. For example, the Golden State Warriors are well-known for their ball movement (passing), which is heavily supported by the data, as the Warriors led the other playoff teams in assist percentage with an average of 67.4% during the seasons they made it to the playoffs within this time span (2017-18, 2018-19, and 2021-22).

Differences in MLR Equation Strength:

Using all of the seasons' data for a single multiple linear regression equation had a lower coefficient of determination value than using multiple linear regression for a single season's data, specifically with the 2018-19 season. In the 2018-19 season, the adjusted R^2 value was about 0.699. This is much higher than the use of the data from all seasons at once, which had an R^2 value of 0.403. In this case, the coefficient of determination indicates that only 40% of the variance in championship likelihood could be determined by a team's number of wins, net rating, and field goal percentage during the regular season using the data from all seasons, making it logical to instead use the 2018-19 model, as it could explain just about 70% of the variance.

Negative Number of Playoff Wins:

Perhaps the most interesting result from the prediction was the negative number of playoff wins predicted for each team. This may have been due to the overall inability for each of the statistics I chose (net rating, wins, and field goal percentage during the regular season) to accurately explain the amount of variance in playoff wins for all teams in each season. However, I supposed I could still utilize the order of the negative number of playoff wins for the prediction.

Confounding Variables:

Unforeseeable Losses:

NBA history demonstrates that wins during the regular season do not always matter. In 2021-22, the Suns won 64 regular season games, the most in the league, but they lost by a substantial 33 points in Game 7 of the Western Conference Semifinals by the Dallas Mavericks, a team that won 52 games that season. In 2023, the Miami Heat won just 44 out of 82 regular season games but made it to the playoffs and still beat the Milwaukee Bucks in the First Round, a team that had 14 more wins than they did, and the Heat even made it to the finals.

Coronavirus Pandemic:

The Coronavirus pandemic may have been the main reason why using the data for all seasons at once for multiple linear regression had a worse adjusted coefficient of determination value than when using the 2018-19 season data. In March of 2020, the regular season was cut short by around 17 total games, leaving most teams playing just 65. Once the season resumed, only the top eight teams in the Eastern and Western conferences were selected to continue the regular season to 72 games for playoff positioning, potentially skewing the results.

Moreover, when the players did continue playing, zero fans were allowed to attend any of the games, which could have caused players to play worse due to the unfamiliar environment. Many players also saw this as an opportunity to rest and have more leisure time than they were able to during the season, which may have affected their physical conditioning and thus performance. Some players refused to take the vaccine, which prevented them from playing in certain arenas where the vaccine was required during the 2020-21 season, which may have affected results.

Different Circumstances:

Another possible reason why using the data for all seasons at once for multiple linear regression had a worse coefficient of determination value than when using the 2017-18 season data is that 2018-19 and 2022-23 are simply two different seasons with different players on different teams in different situations. Since my model was based off of the data gathered from the 2018-19, it is possible that the data is biased toward teams that were either close to winning the championship or won the championship in 2018-19 in the Milwaukee Bucks, Toronto Raptors, and Philadelphia 76ers.

Injuries:

When looking at the preliminary prediction, injuries may have played a significant factor in selecting the Bucks rather than the Nuggets. For example, in April 2021, Denver star point guard Jamal Murray suffered an Anterior Cruciate Ligament (ACL) tear that sidelined him during the entire 2021-22 season. However, he recovered and returned by the start of the 2022-23 season, and the Nuggets won the championship. Since Murray has a dramatic impact on the team being one of their stars, it is possible that he was a big contributor to their recent playoff success.

Reflection:

Strengths:

Strength	Significance of Strength
Simple Interpretation	Multiple linear regression provides multiple slope coefficients that are easily interpretable for each independent variable, which makes it easy to understand the magnitude of each independent variable in the context of predicting championship likelihood.
Quantitative Analysis	Using multiple linear regression provides a quantitative framework for analyzing the relationship between regular season statistics and future playoff success, rather than perhaps using qualitative data from NBA analysts, offering a more objective prediction.
Use of Historical Data	Using past regular season data from the NBA may be valuable as previous seasons are able to create patterns that can be used to determine what may happen during a future season.

Weaknesses:

Weakness	Significance of Weakness
Linearity Assumption	Multiple linear regression, like I previously mentioned, assumes that there will be a linear relationship between the independent variables (regular season statistics) and the dependent variable (playoff wins), which may or may not always be true, thus potentially reducing the accuracy of prediction.
Outliers	Outliers were not accounted for when creating the multiple linear regression, which may have affected the model's overall fit to the data and thus may have skewed the prediction.
Seasonal Variations	NBA seasons can differ in multiple ways each year, such as differences in team dynamics, player performance, rule changes, and others. Using regular season statistics for a set period of time (in my case, five seasons) may not be able to accurately predict the circumstances for the next season.

Conclusion:

My mathematical investigation has raised several questions, such as, "What regular season factors contribute most to playoff success present day in basketball considering most of these factors showed a decline in significance? Why have they declined in significance?". Additionally, it raises the question of the potential implication of prediction models like this for NBA teams, and how we can create better tools for predicting the success of teams. I previously mentioned that users have moved to machine learning for predicting playoff success, but other than finding out what team may or may not win the championship that year, what other things could these models be used for? Perhaps NBA teams could use these models to predict their own future playoff success and make business decisions, such as trading players, based on those predictions to create the best possible chance for them to succeed.

To conclude, my research found that the three most important regular season factors that contribute to playoff success in order are net rating, wins, and field goal percentage, all during the regular season. I used a total of just five factors, the other two being assist percentage and three-pointers made during the regular season, which does create a limited conclusion. Though these statistics are fairly basic, I was able to use them to predict the 2024 NBA champion; my answer being the Milwaukee Bucks.

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